

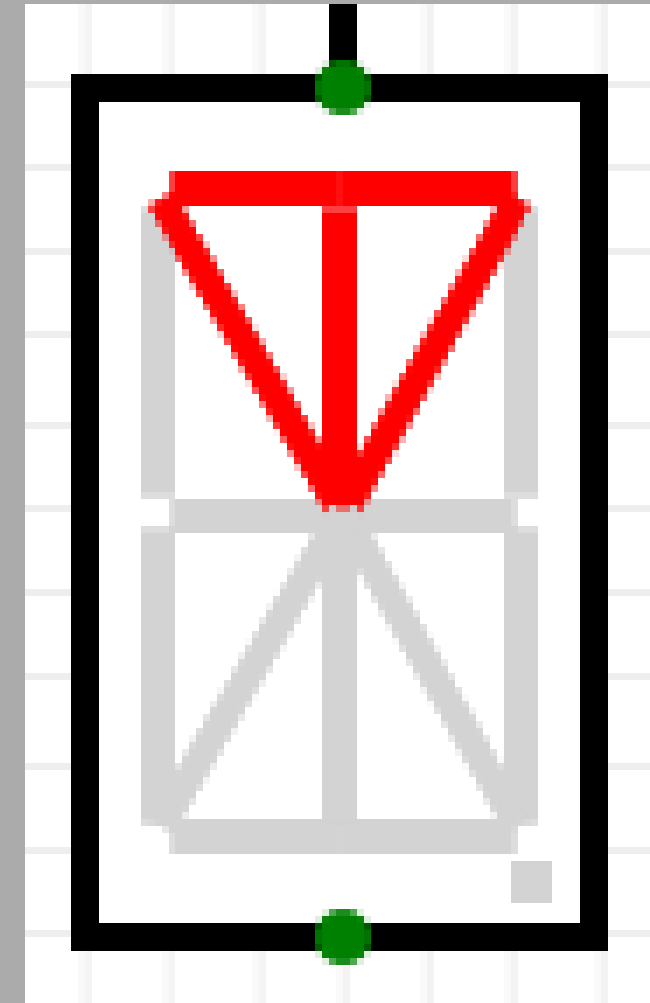
2026.1

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SISTEMAS DIGITAIS

PROPOSTA

Nosso projeto teve como objetivo projetar e simular um circuito de 16 caracteres que mostra a frase : "EU-<3-Pernambuco-"



DESENVOLVIMENTO

TABELA-VERDADE

	A	B	C	D	a1	a2	b	c	d1	d2	e	f	g1	g2	h	i	j	k	l	m	
E	0	0	0	0	0	1	1	0	0	1	1	1	1	1	1	0	0	0	0	0	0
U	0	0	0	0	1	0	0	1	1	1	1	1	1	0	0	0	0	0	0	0	0
-	0	0	1	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0
<3	0	0	1	1	1	1	1	0	0	0	0	0	0	0	0	1	1	1	0	0	0
-	0	1	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0
P	0	1	0	1	1	1	1	1	0	0	0	1	1	1	1	0	0	0	0	0	0
E	0	1	1	0	1	1	1	0	0	1	1	1	1	1	1	0	0	0	0	0	0
R	0	1	1	1	1	1	1	0	0	0	0	1	1	0	0	0	0	1	0	0	1
N	1	0	0	0	0	0	0	1	1	0	0	1	1	0	0	1	0	0	0	0	1
A	1	0	0	1	1	1	1	1	1	0	0	1	1	1	1	0	0	0	0	0	0
M	1	0	1	0	1	1	1	1	1	0	0	1	1	0	0	0	1	0	0	1	0
B	1	0	1	1	0	0	0	0	1	1	1	1	1	1	1	0	0	0	0	0	0
U	1	1	0	0	0	0	0	1	1	1	1	1	1	0	0	0	0	0	0	0	0
C	1	1	0	1	1	1	1	0	0	1	1	1	1	0	0	0	0	0	0	0	0
O	1	1	1	0	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
-	1	1	1	1	1	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0

LINK PARA EXCEL

DESENVOLVIMENTO

MAPAS KARNAUGH

SIMPLIFICAÇÃO

a1

	-C		C		
-A	1	0	1	0	-B
	0	1	1	1	B
A	0	1	0	1	B
	0	1	0	1	-B
	-D	D	-D		

a2

	-C		C		
-A	1	0	1	0	-B
	0	1	1	1	B
A	0	1	0	1	B
	0	1	0	1	-B
	-D	D	-D		

$$a1 = a2 = \neg A \neg B (C \odot D) + B(\neg AD + C \neg D) + A(C \oplus D)$$

b

	-C		C		
-A	0	1	0	0	-B
	0	1	0	0	B
A	1	0	0	1	B
	1	1	0	1	-B
	-D	D	-D		

c

	-C		C		
-A	0	1	0	0	-B
	0	0	0	0	B
A	1	0	0	1	B
	1	1	1	1	-B
	-D	D	-D		

$$b = A(\neg B \neg C + \neg D) + (\neg A \neg CD)$$

$$c = A(\neg B + \neg D) + \neg A \neg B \neg CD$$

d1

	-C		C		
-A	1	1	0	0	-B
	0	0	0	1	B
A	1	1	0	1	B
	0	0	1	0	-B
	-D	D	-D		

d2

	-C		C		
-A	1	1	0	0	-B
	0	0	0	1	B
A	1	1	0	1	B
	0	0	1	0	-B
	-D	D	-D		

$$d1 = d2 = \neg C(A \odot B) + C(B \neg D + A \neg BD)$$

e

	-C		C		
-A	1	1	0	0	-B
	0	1	1	1	B
A	1	1	0	1	B
	1	1	1	1	-B
	-D	D	-D		

f

	-C		C		
-A	1	1	0	0	-B
	0	1	1	1	B
A	1	1	0	1	B
	1	1	1	1	-B
	-D	D	-D		

DESENVOLVIMENTO

MAPAS KARNAUGH

SIMPLIFICAÇÃO

1

	-C		C		
-A	1	0	0	1	-B
	1	1	0	1	B
A	0	0	1	0	-B
	0	1	1	0	B
	-D	D	-D	D	

g2

	-C		C		
-A	1	0	0	1	-B
	1	1	0	1	B
A	0	0	1	0	-B
	0	1	1	0	B
	-D	D	-D	D	

$$g1 = g2 = -A(-D + B - C) + AD(-B + C)$$

	-C		C		
-A	0	0	1	0	-B
	0	0	0	0	B
A	0	0	0	0	-B
	1	0	0	0	B
	-D	D	-D	D	

i

	-C		C		
-A	0	0	1	0	-B
	0	0	0	0	B
A	0	0	0	0	-B
	0	0	0	1	B
	-D	D	-D	D	

$$h = -B(A - C - D + -ACD)$$

$$i = -BC(A \oplus D)$$

	-C		C		
-A	0	0	1	0	-B
	0	0	1	0	B
A	0	0	0	0	-B
	0	0	0	0	B
	-D	D	-D	D	

k

	-C		C		
-A	0	0	0	0	-B
	0	0	0	0	B
A	0	0	0	0	-B
	0	0	0	0	B
	-D	D	-D	D	

$$j = -ACD$$

$$k = 0$$

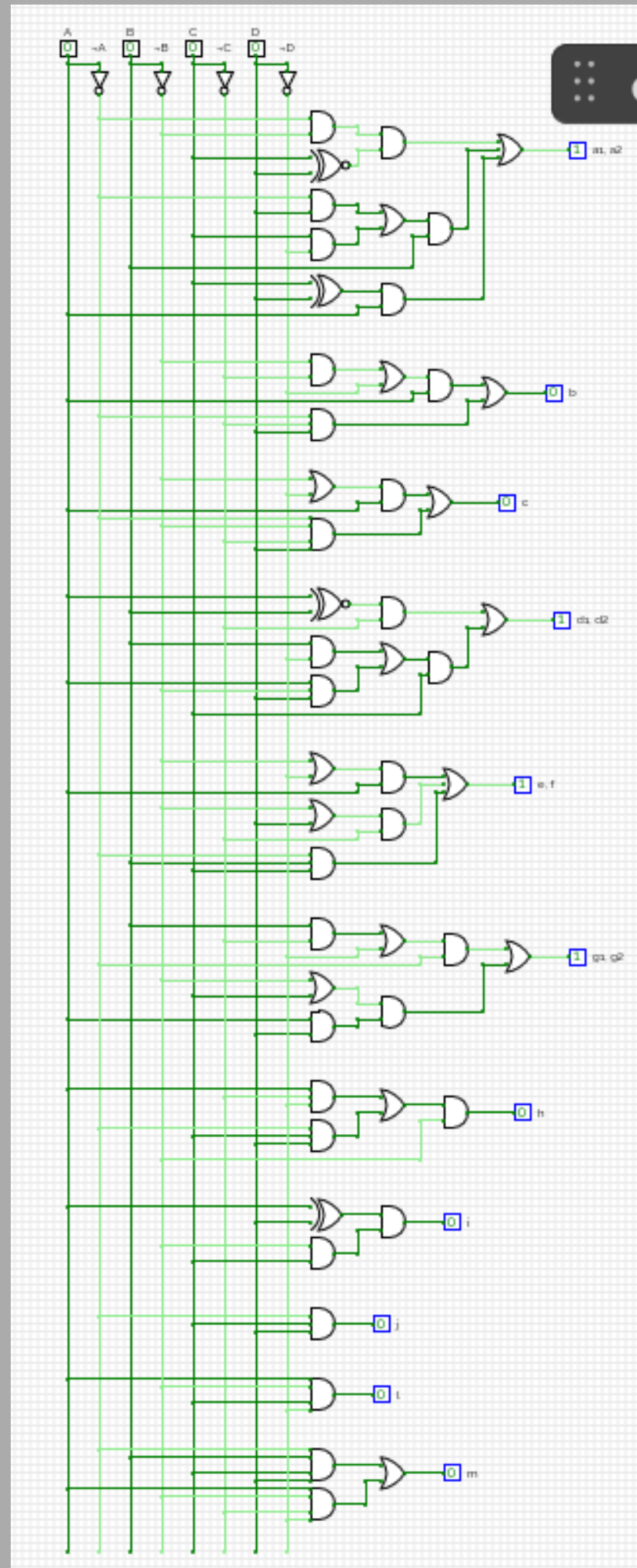
	-C		C		
-A	0	0	0	0	-B
	0	0	0	0	B
A	0	0	0	0	-B
	0	0	0	1	B
	-D	D	-D	D	

m

	-C		C		
-A	0	0	0	0	-B
	0	0	1	0	B
A	0	0	0	0	-B
	1	0	0	0	B
	-D	D	-D	D	

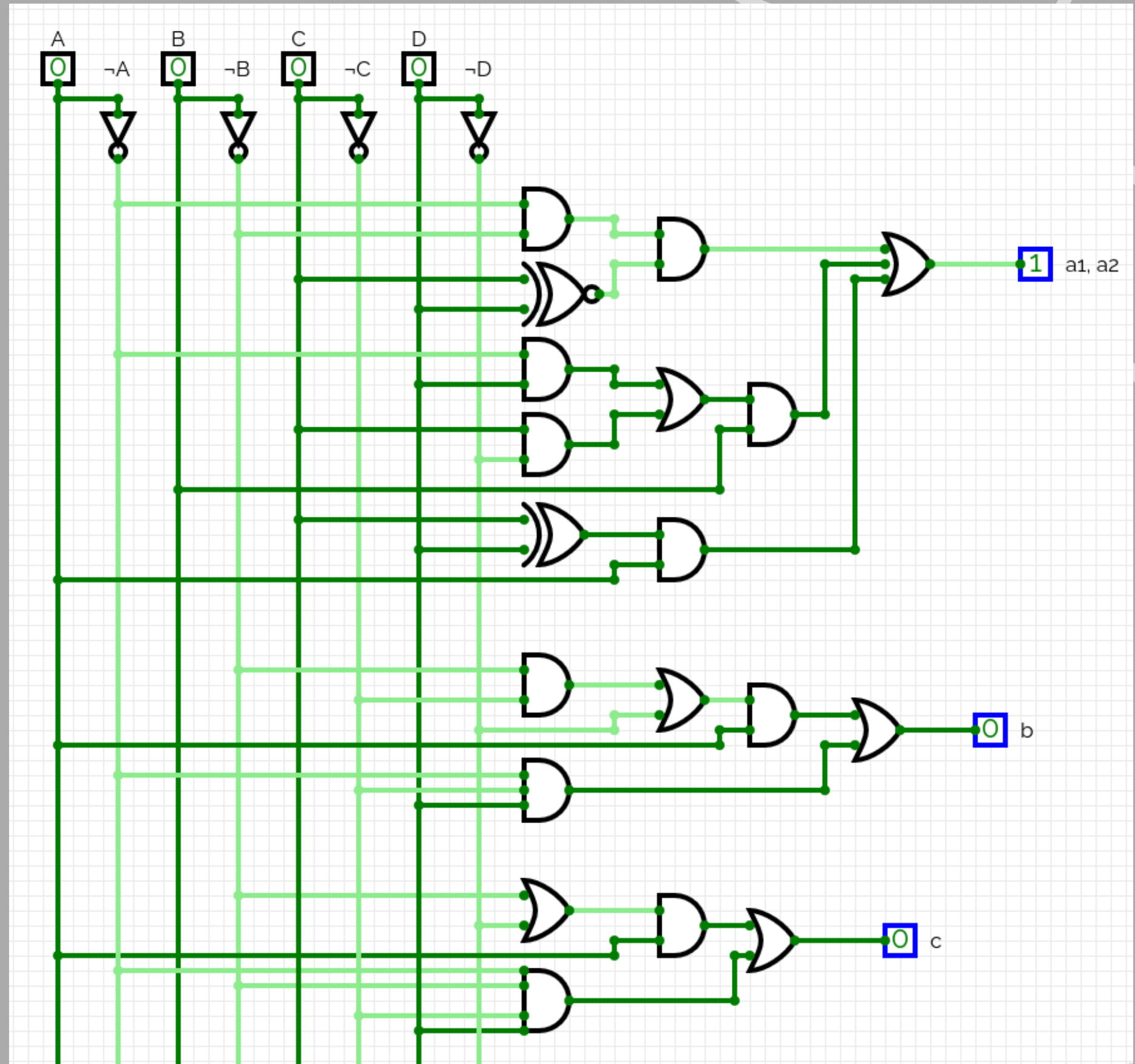
DESENVOLVIMENTO

CIRCUITO SEGMENTOS



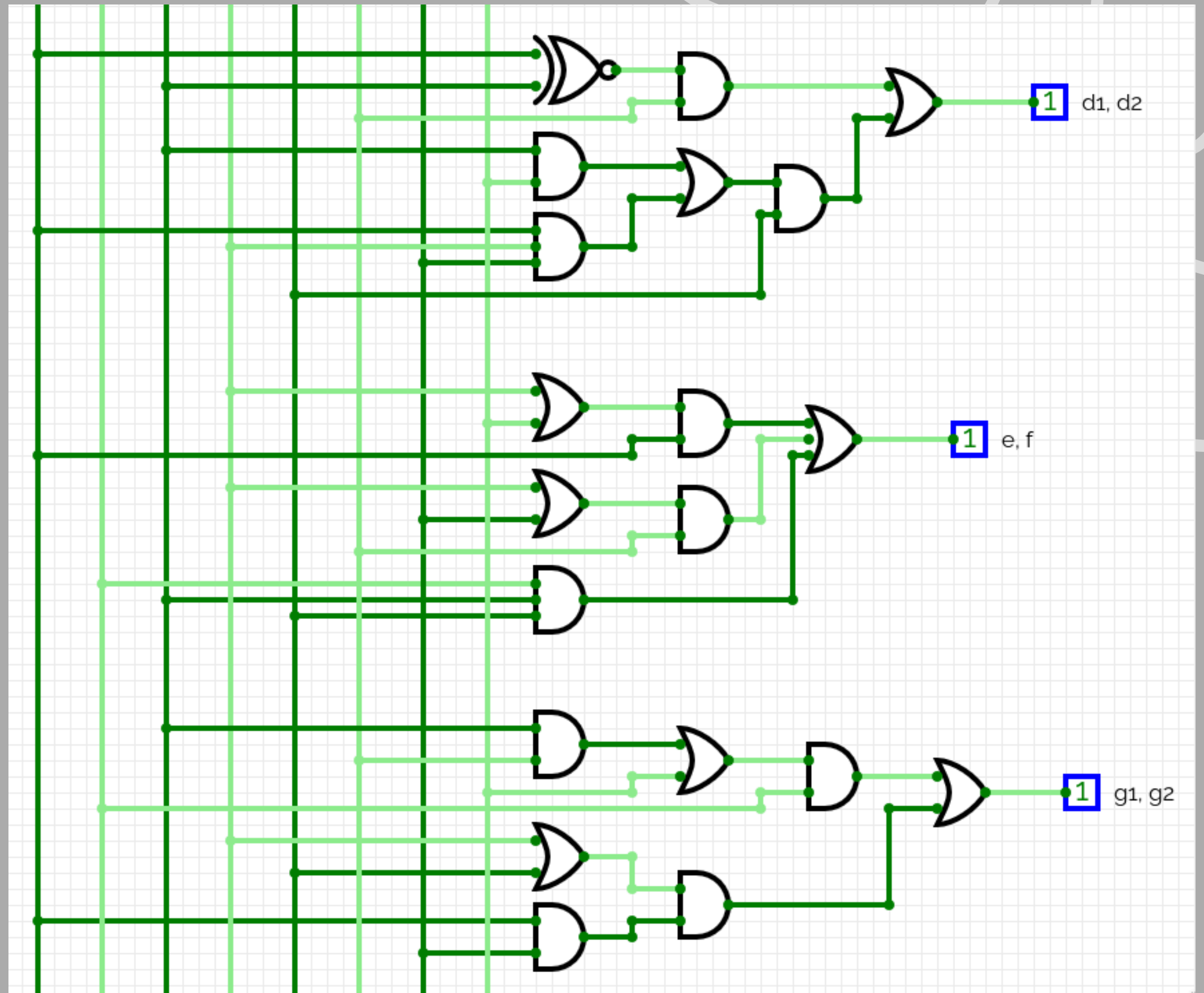
DESENVOLVIMENTO

CIRCUITO SEGMENTOS



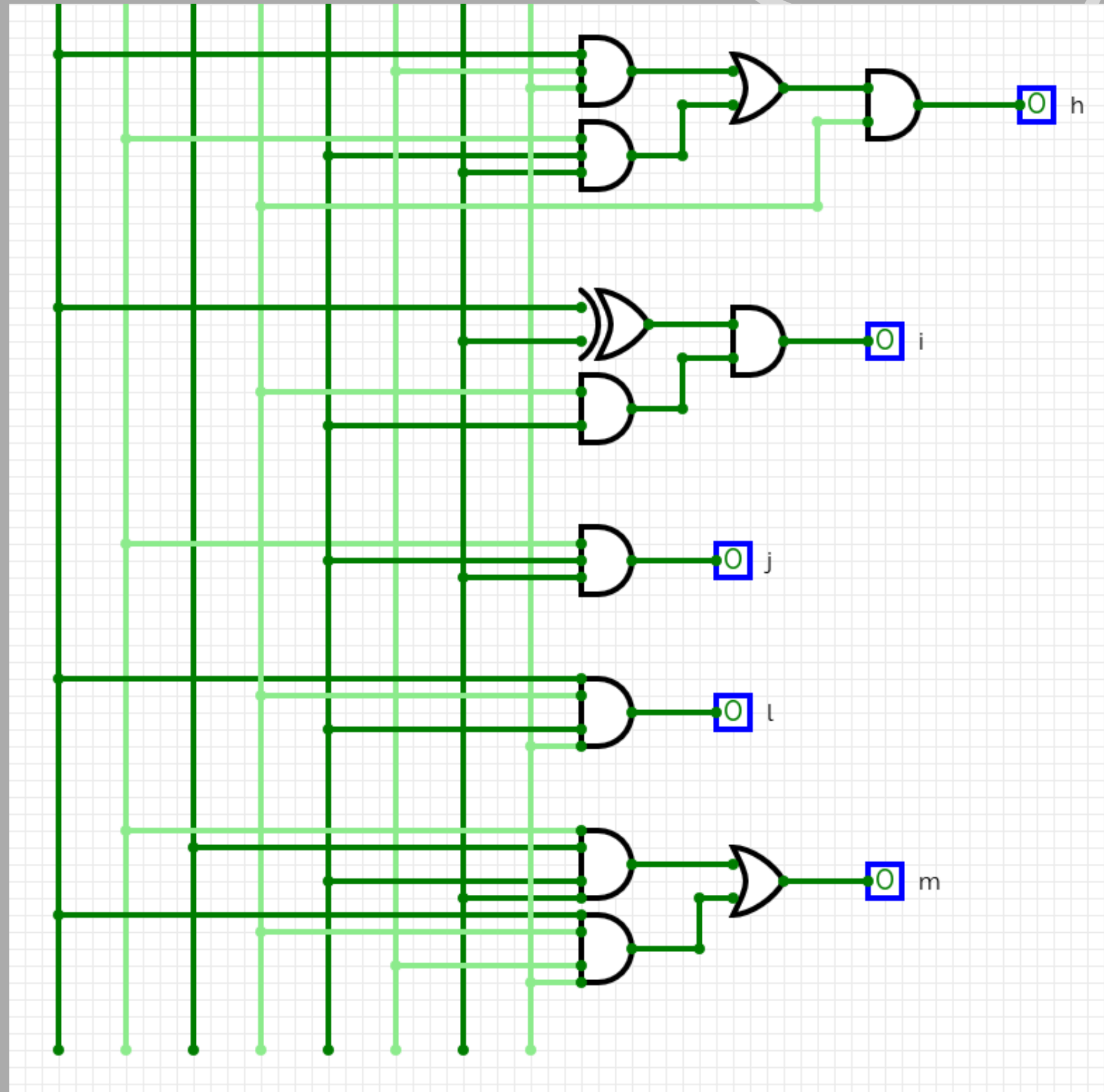
DESENVOLVIMENTO

CIRCUITO SEGMENTOS



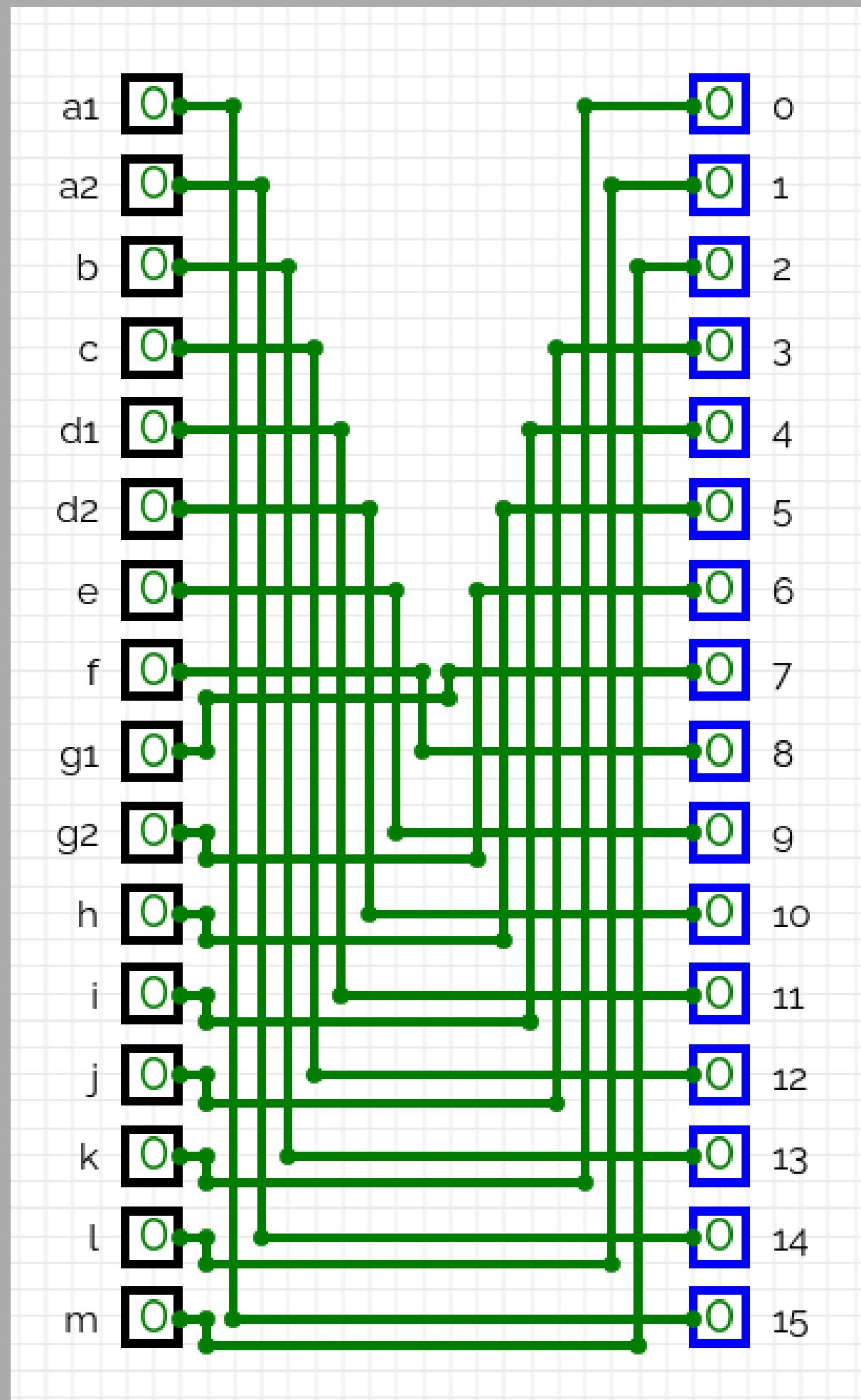
DESENVOLVIMENTO

CIRCUITO SEGMENTOS



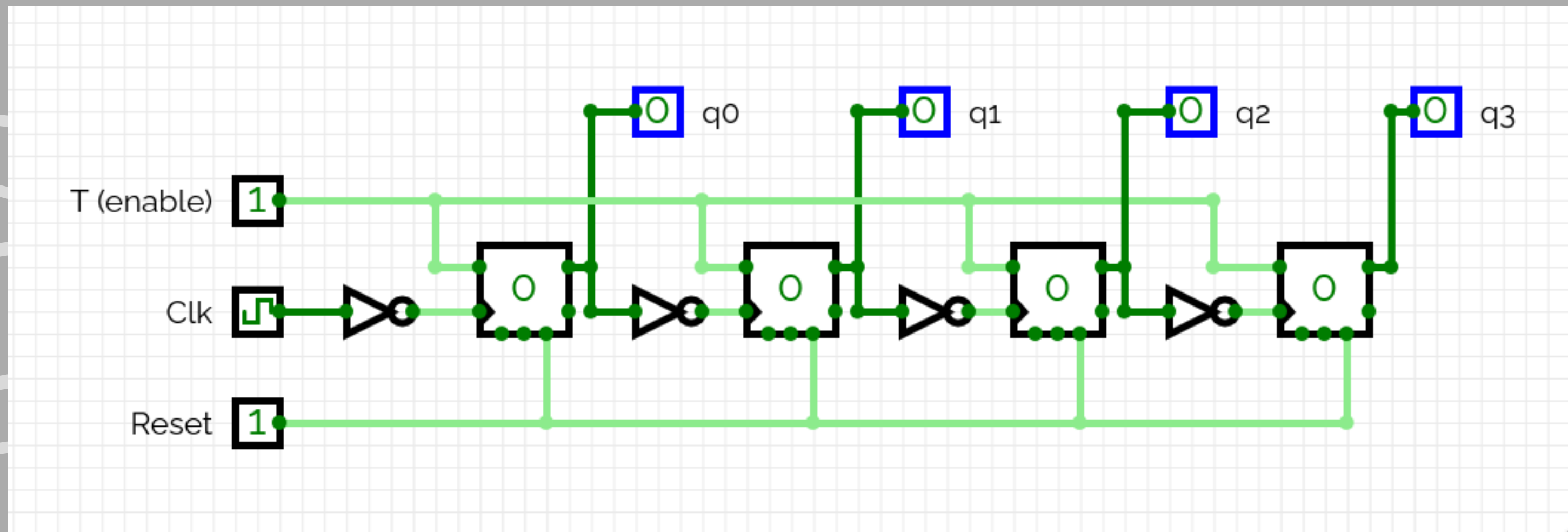
DESENVOLVIMENTO

CAMADA CONVERSÃO



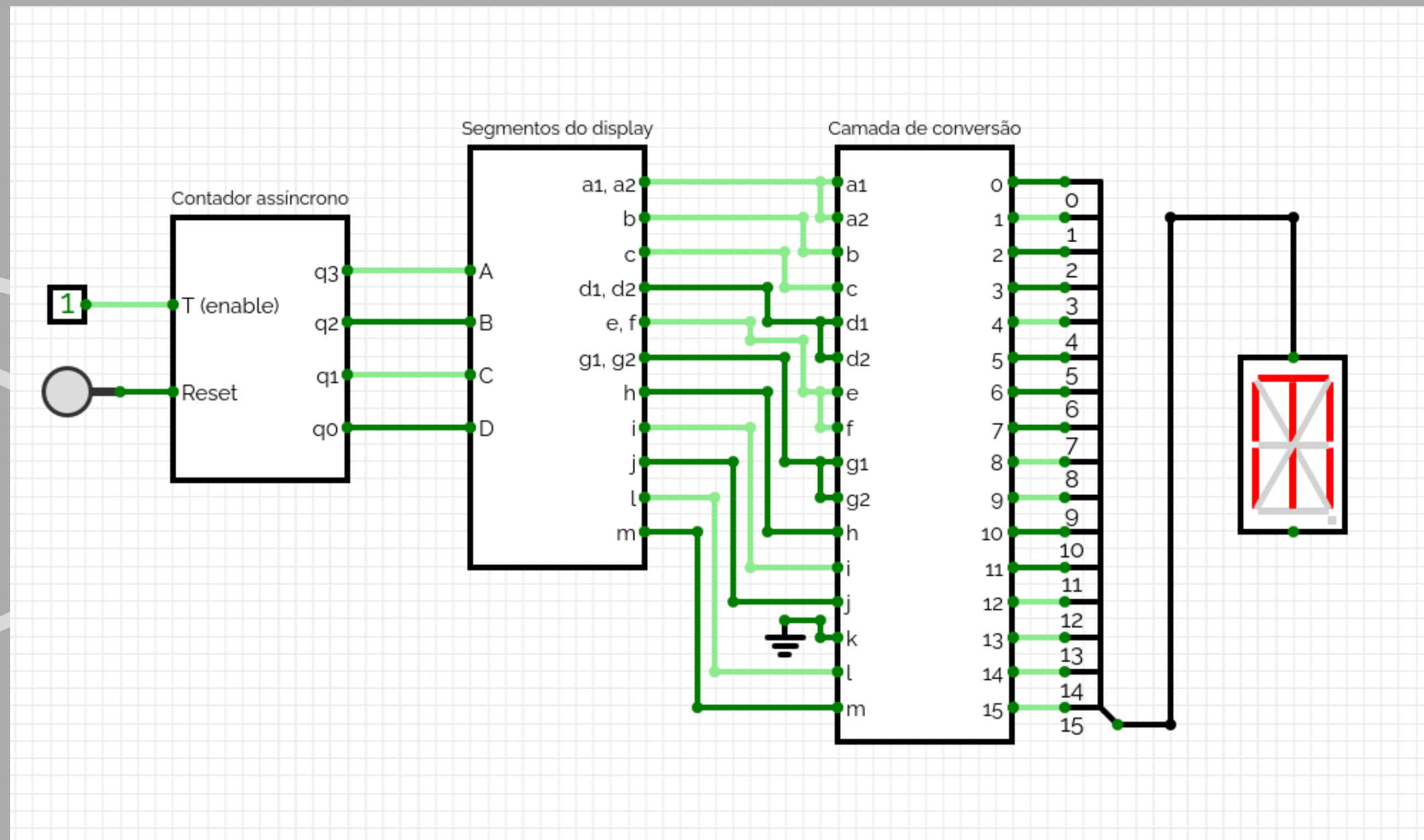
DESENVOLVIMENTO

CONTADOR ASSÍNCRONO



DESENVOLVIMENTO

CIRCUITO FINAL



LINK

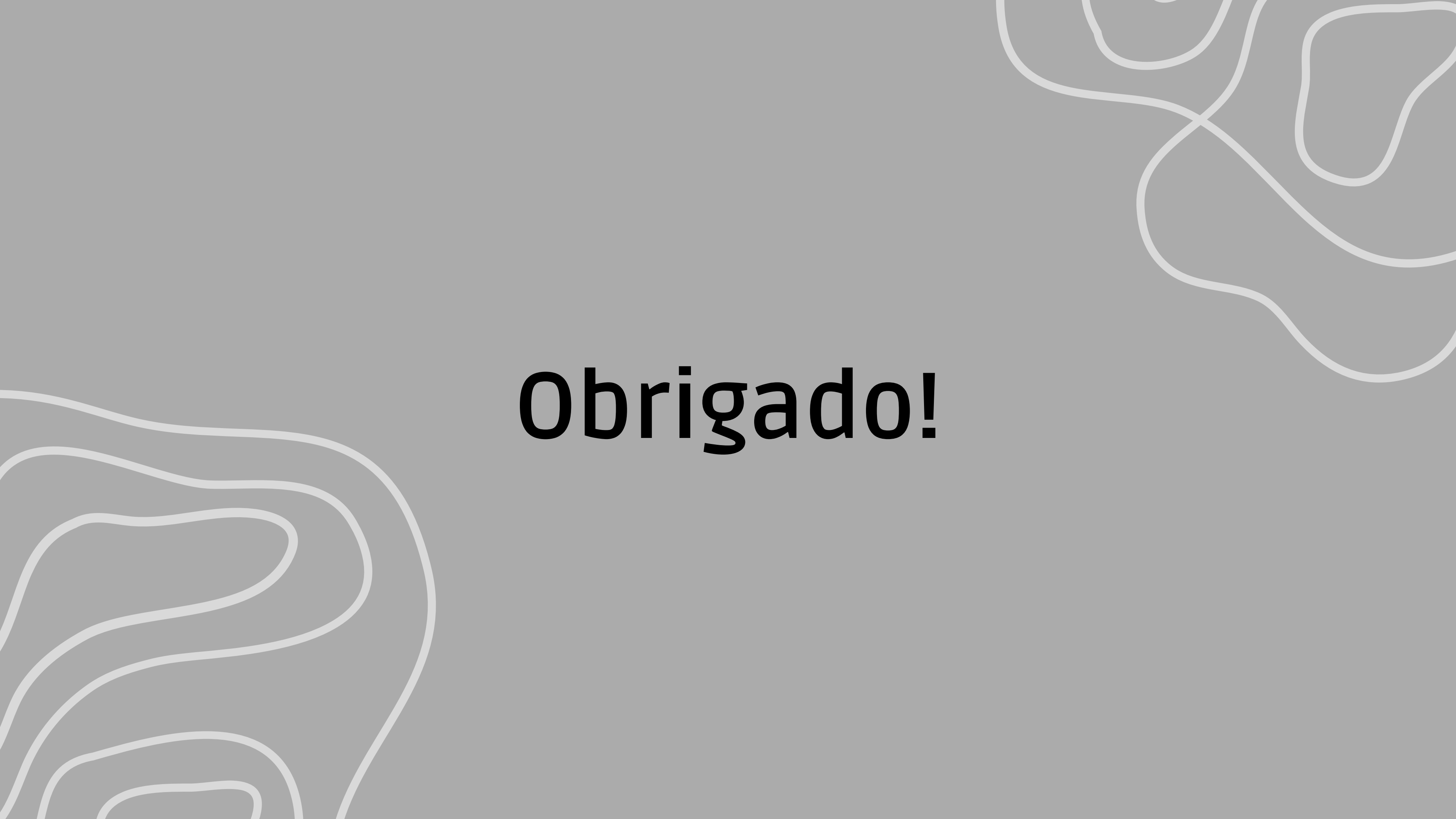
VIDEO DEMONSTRATIVO

LINK

REFERÊNCIAS

TOCCI, Ronald J.; WIDMER, Neal S. Sistemas digitais: princípios e aplicações. 8. ed. São Paulo: Prentice-Hall, 2003.

IDOETA, Ivan Valeije; CAPUANO, Francisco Gabriel. Elementos de eletrônica digital. 32. ed. São Paulo: Érica, 2001.

The image features a solid light gray background. In the top right and bottom left corners, there are abstract, hand-drawn white line patterns. These lines are fluid and organic, resembling stylized waves or calligraphic flourishes. The central text 'Obrigado!' is positioned in the middle of the frame, rendered in a bold, black, sans-serif typeface.

Obrigado!